Lecture 4: Regularized Regression - Lasso

Max G'Sell
Machine Learning I: 46-926

November 8, 2020

Announcements

- First quiz was turned in yesterday. How did that work out?

Today: Finishing basics, Linear Regression and Regularization

- Extending linear regression to work in high dimensions
  - Lasso
  - Model selection, cross-validation (unlikely)

Review: Supervised learning

Observe $(X_1, Y_1), \ldots (X_n, Y_n) \overset{iid}{\sim} \mathcal{P}$. Estimate $\hat{\mu}(x)$ so that $\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X))$ is small for new $(X, Y) \sim \mathcal{P}$.

If we are minimizing expected squared prediction error:

- If we actually knew $\mathcal{P}$, the best we could do is

$$\mu(x) = \mathbb{E}(Y|X = x)$$

- If we are estimating $\hat{\mu}$, then

$$\mathbb{E}\left(\mathcal{L}(Y, \hat{\mu}(X)) \mid X = x\right) = \sigma_x^2 + \text{bias}(\hat{\mu}(x))^2 + \text{var}(\hat{\mu}(x))$$

Review: Linear regression

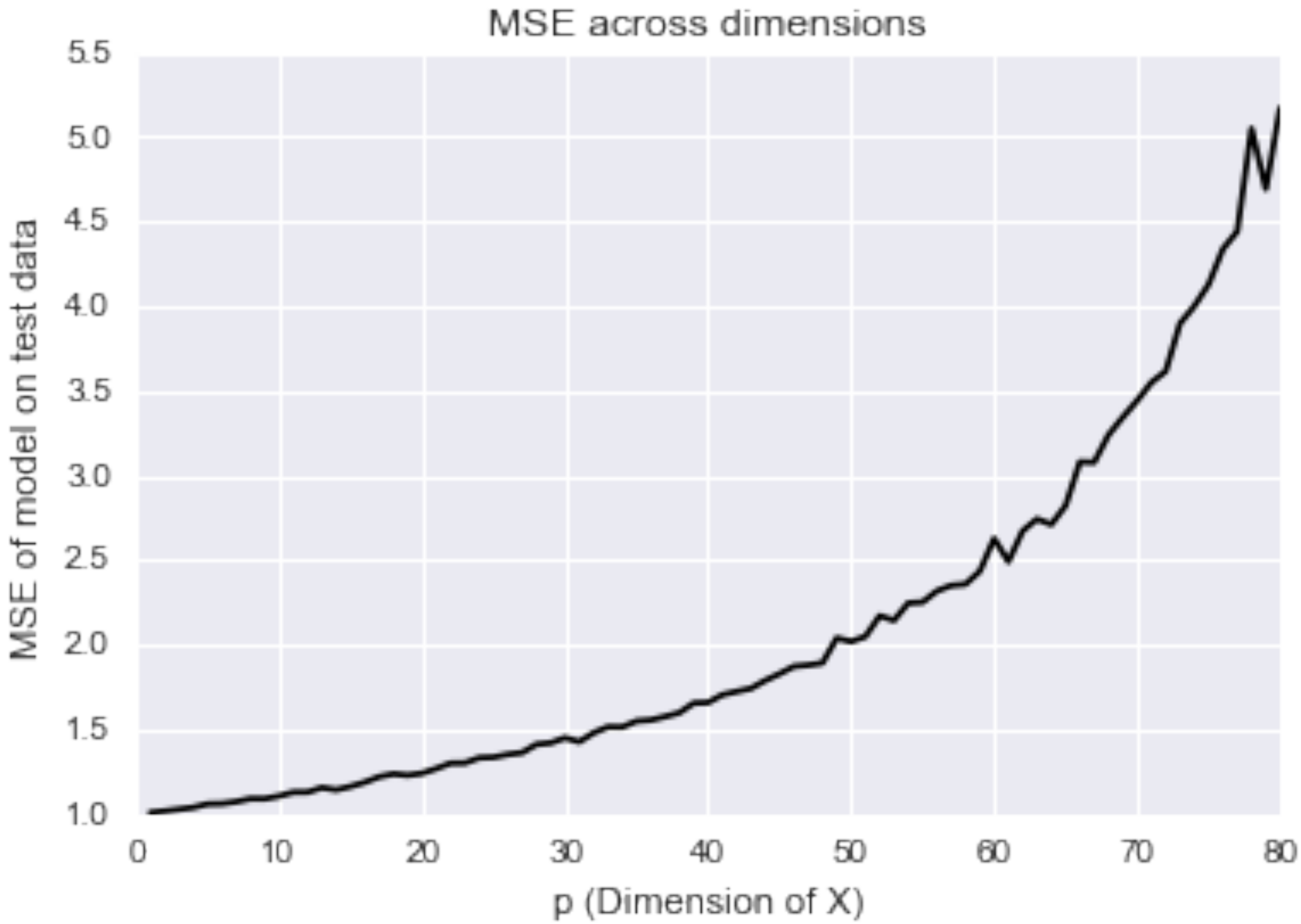
We can think of linear regression as one way to approximate $\mathbb{E}(Y|X = x)$ across x values, imagining:

$$\mathbb{E}(Y|X = x) \approx x^T\beta$$

This can work well if the linear model is not a bad approximation.

However, we noticed that, as the number of variables, p, grows large, the variance of linear regression grows rapidly. This leads to poor predictions even with the linear model is true!

Linear regression in high dimensions



This is a simulation from a true linear model with independent Gaussian errors – the dream! (Also the plot from your homework.)

Ridge regression

Ridge regression is like least squares but shrinks the estimated coefficients towards zero. Given a response vector $y \in \mathbb{R}^n$ and a predictor matrix $X \in \mathbb{R}^{n \times p}$, the ridge regression coefficients are defined as

$$\hat{\beta}^{\text{ridge}} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2$$
$$= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}}$$

Ridge regression

$$\underbrace{\sum_{i=1}^{n} (y_i - \beta_0)^2}_{\bar{Y}} \qquad \underbrace{\| y - \beta_0 - \cancel{X\beta} \|_2^2}_{} + \lambda \|\beta\|_2^2$$

$$\hat{\beta}^{\text{ridge}} = \underset{\beta \in \mathbb{R}^p}{\text{argmin}} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$\|y\|_2^2 + \beta = 0$$

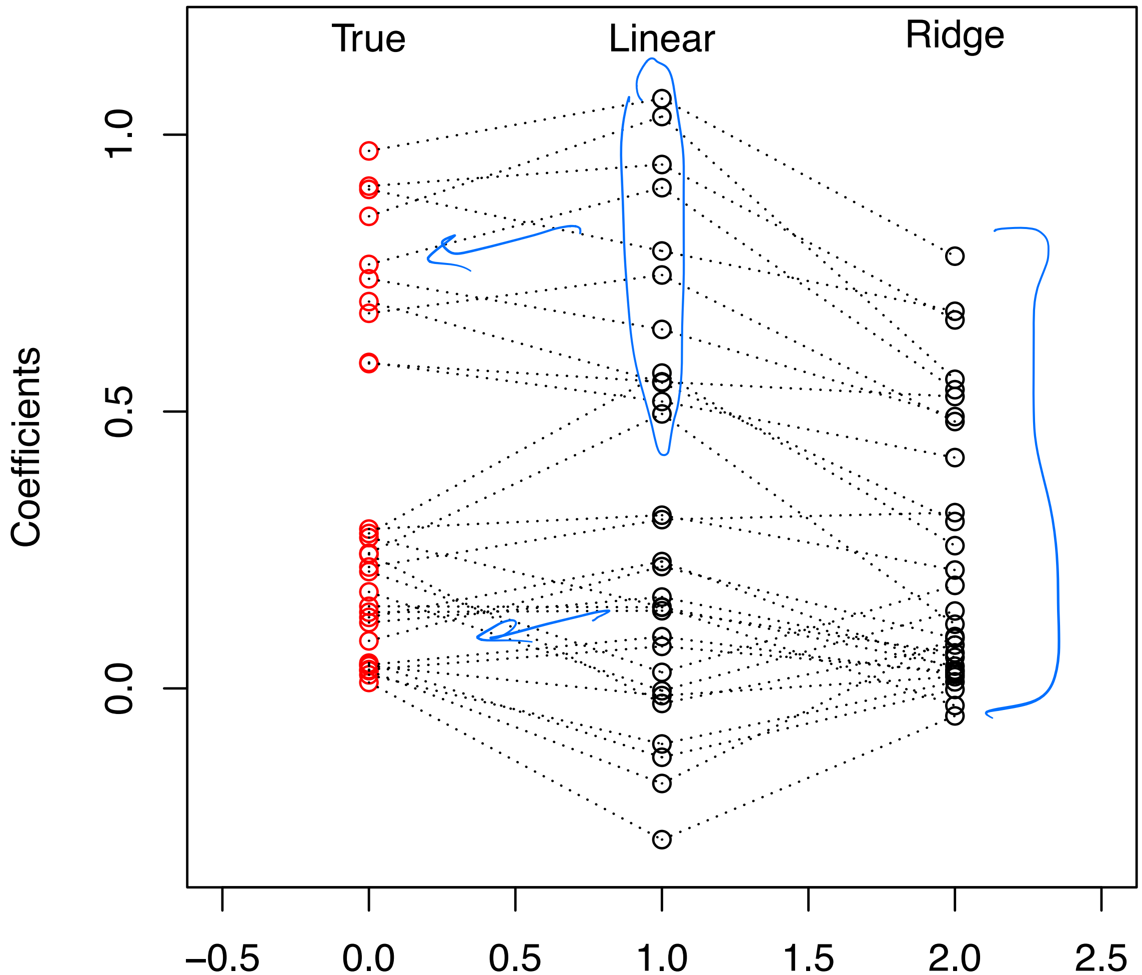
$$= \underset{\beta \in \mathbb{R}^p}{\text{argmin}} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}}$$

Here $\lambda \geq 0$ is a tuning parameter, which controls the strength of the penalty term. Note that:

- When $\lambda = 0$, LS
- When $\lambda \to \infty$, $\hat{\beta} = 0$, $\bar{Y}$
- For λ in between, we are balancing two ideas: fitting a linear model of y on X, and shrinking the coefficients

Example: visual representation of ridge coefficients

A visual representation of the ridge regression coefficients for the same example ($n = 50$, $p = 30$, and $\sigma^2 = 1$; 10 large true coefficients, 20 small) at $\lambda = 25$:



We see that squishing the coefficients toward zero gives:

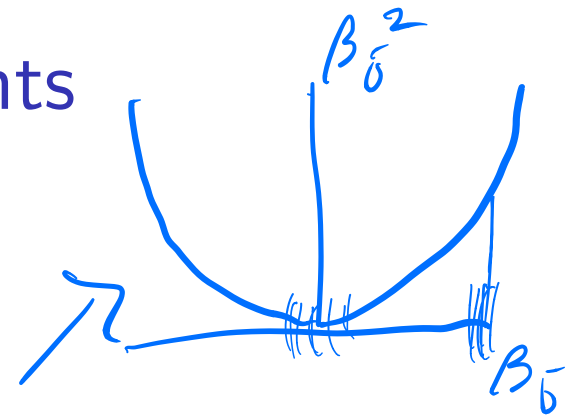
- A decrease in variance!
- A decrease in MSE!
- An increase in bias. . .

However, ridge regression:

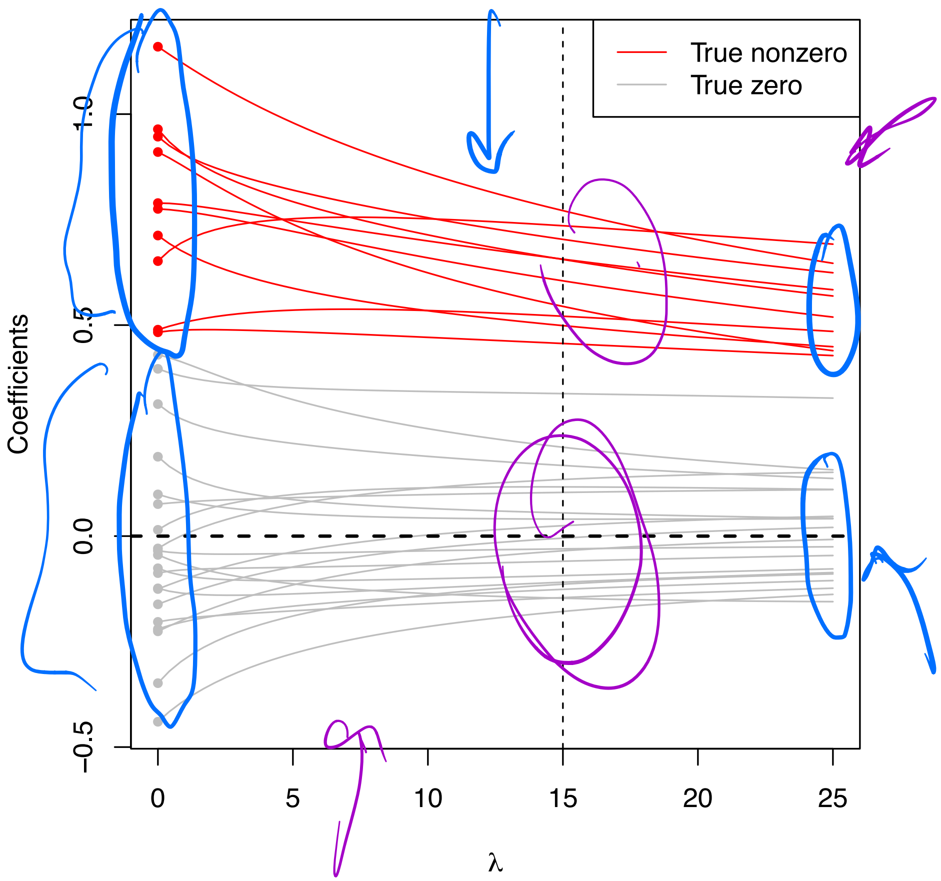
- Seems quite harsh on large, important coefficients.

- Leaves us with a large linear model, which is not that interpretable when we have many variables.

Ridge regression coefficients



Ridge regression coefficients, plotted against λ:

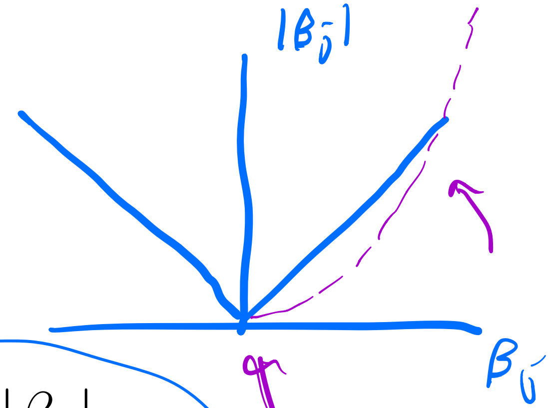


The red paths correspond to the true nonzero coefficients; the gray paths correspond to true zeros. The vertical dashed line at $\lambda = 15$ marks the point above which ridge regression's MSE starts losing to that of linear regression

An important thing to notice is that the gray coefficient paths are not exactly zero; they are shrunken, but still nonzero

The lasso

$\|v\|_p = \left(\sum_i |v_i|^p\right)^{1/p}$ $\quad \|v\|_1 = \sum |v_i|$



The lasso[1] estimate is defined as

$$\hat{\beta}^{\text{lasso}} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

$$= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_1}_{\text{Penalty}}$$

The squared ℓ_2 penalty $\|\beta\|_2^2$ of ridge regression, has been replaced by an ℓ_1 penalty $\|\beta\|_1$. Even though these problems look similar, their solutions behave very differently

Note the name "lasso" is actually an acronym for: Least Absolute Selection and Shrinkage Operator

$\lambda\left(\alpha \|\beta\|_1 + (1-\alpha)\|\beta\|_2^2\right) \leftarrow$ Elastic Net

[1] Tibshirani (1996), "Regression Shrinkage and Selection via the Lasso"

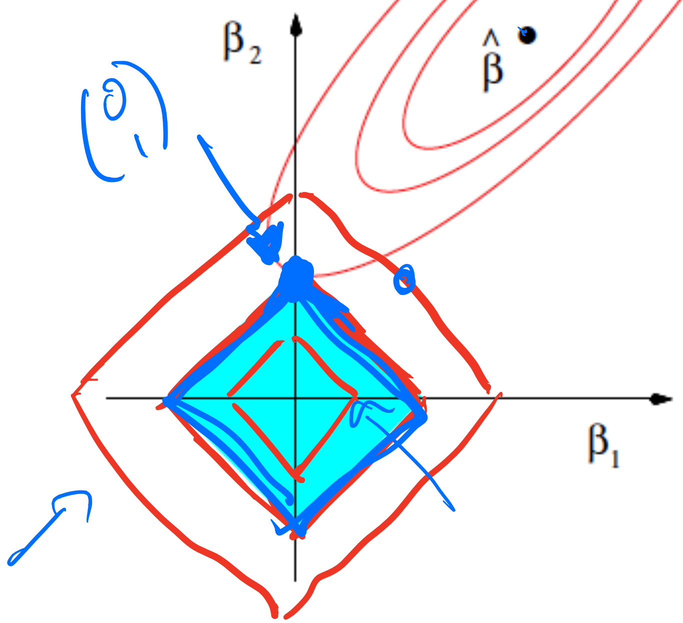
$$\|Y - XB\|_2^2 + \lambda \|\beta\|_1$$

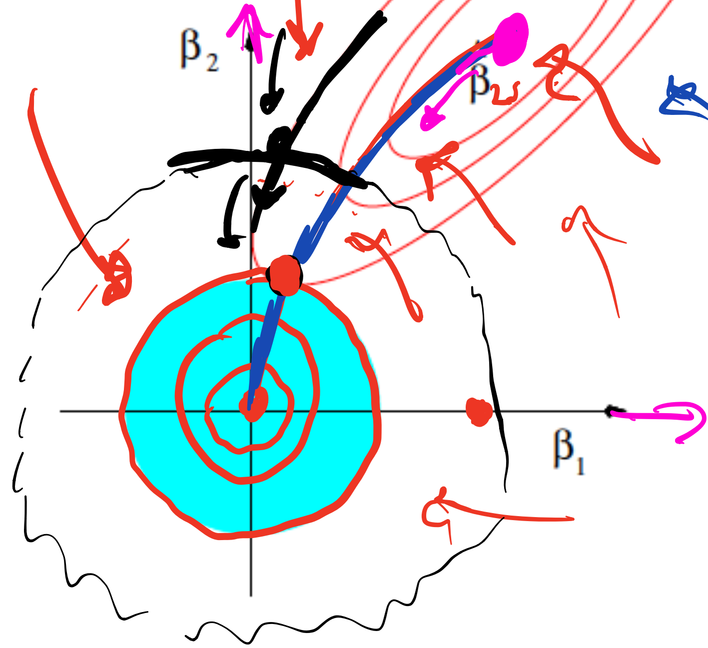
$$(|\beta_1| + |\beta_2|)\lambda$$

$$|\beta_1| + |\beta_2| = 1$$

$$\|Y - XB\|_2^2 + \cancel{\lambda}\, \|\beta\|_2^2$$

$$\sum \beta_i^2$$





$$\|Y - XB\|_2^2 = (Y - XB)^T (Y - XB) = Y^T Y - 2 Y^T X B + B^T X^T X B$$

$\sum|\beta_j|$

$$\hat{\beta}^{\text{lasso}} = \underset{\beta\in\mathbb{R}^p}{\operatorname{argmin}}\ \underbrace{\|y - X\beta\|_2^2} + \lambda\underbrace{\|\beta\|_1}_{\sum|\beta_j|}$$

The tuning parameter λ controls the strength of the penalty.

- If $\lambda = 0$: $\hat{\beta}_{LS}$ if exists.
- As $\lambda \to \infty$: $\hat{\beta} = 0$

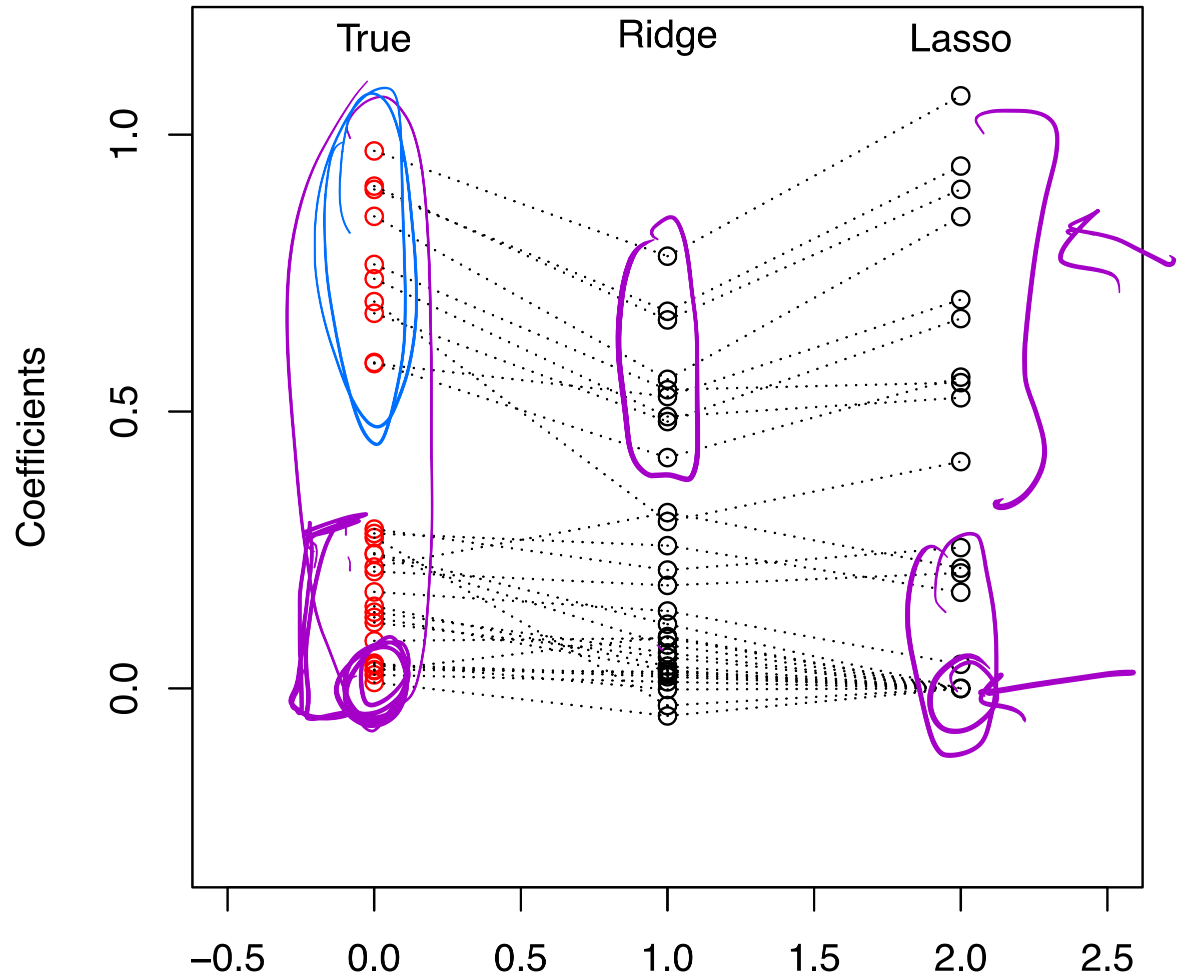
For λ in between these two extremes, we are balancing two ideas: fitting a linear model of y on X, and shrinking the coefficients. But the nature of the ℓ_1 penalty causes some coefficients to be shrunken to zero exactly

$$\hat{\beta} = \begin{array}{cl} \begin{pmatrix} 2.3 \\ 0 \\ 0 \\ 1.2 \\ 0 \end{pmatrix} & \begin{array}{l} \leftarrow x_1 \\ \\ \\ \leftarrow x_4 \\ \\ \end{array} \end{array}$$

This leads to variable selection!

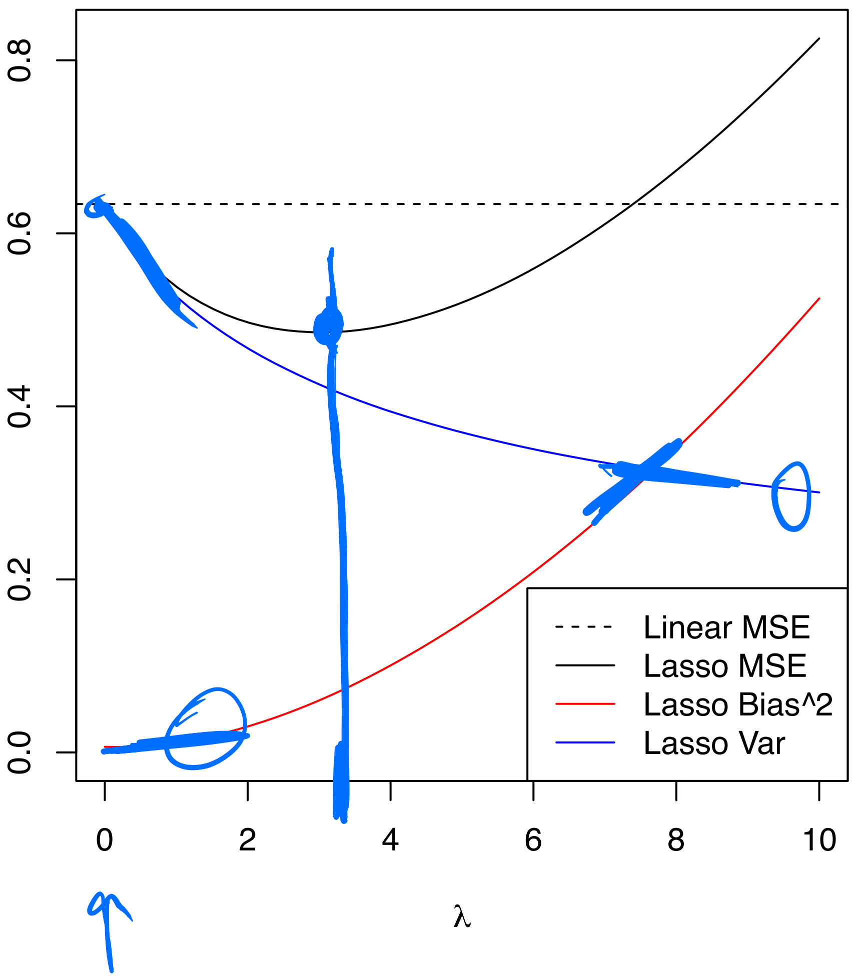
Example: visual representation of lasso coefficients

Our running example from last time: $n = 50$, $p = 30$, $\sigma^2 = 1$, 10 large true coefficients, 20 small. Here is a visual representation of lasso vs. ridge coefficients (with the same degrees of freedom):



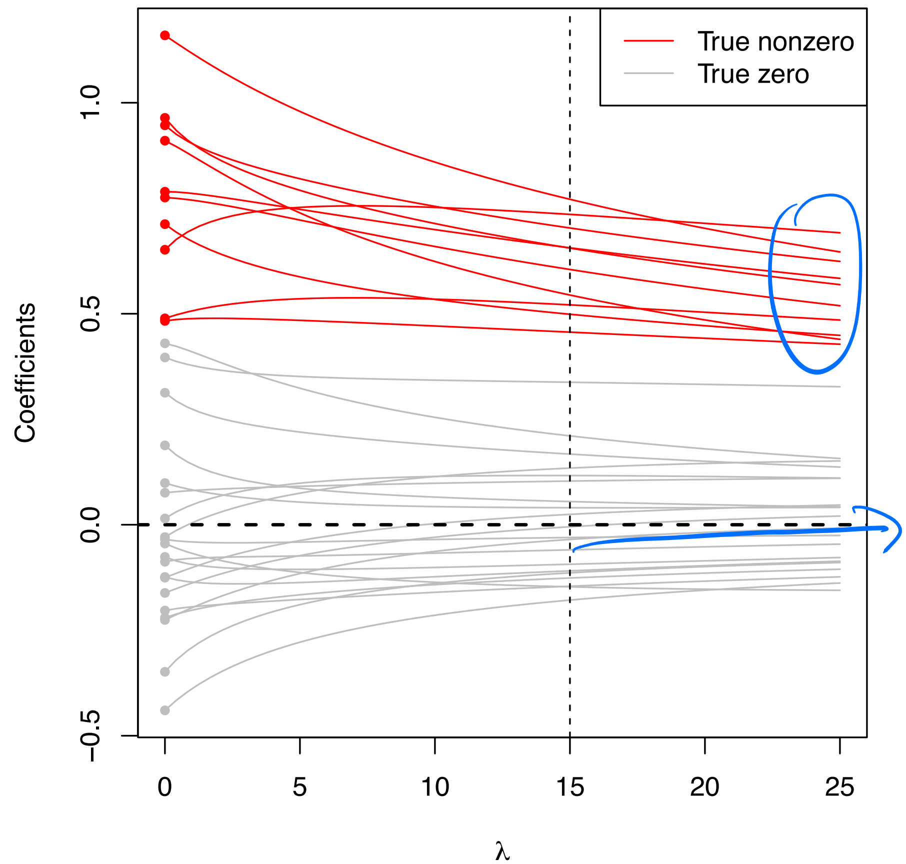
Example: subset of small coefficients

Example: $n = 50$, $p = 30$; true coefficients: 10 large, 20 small



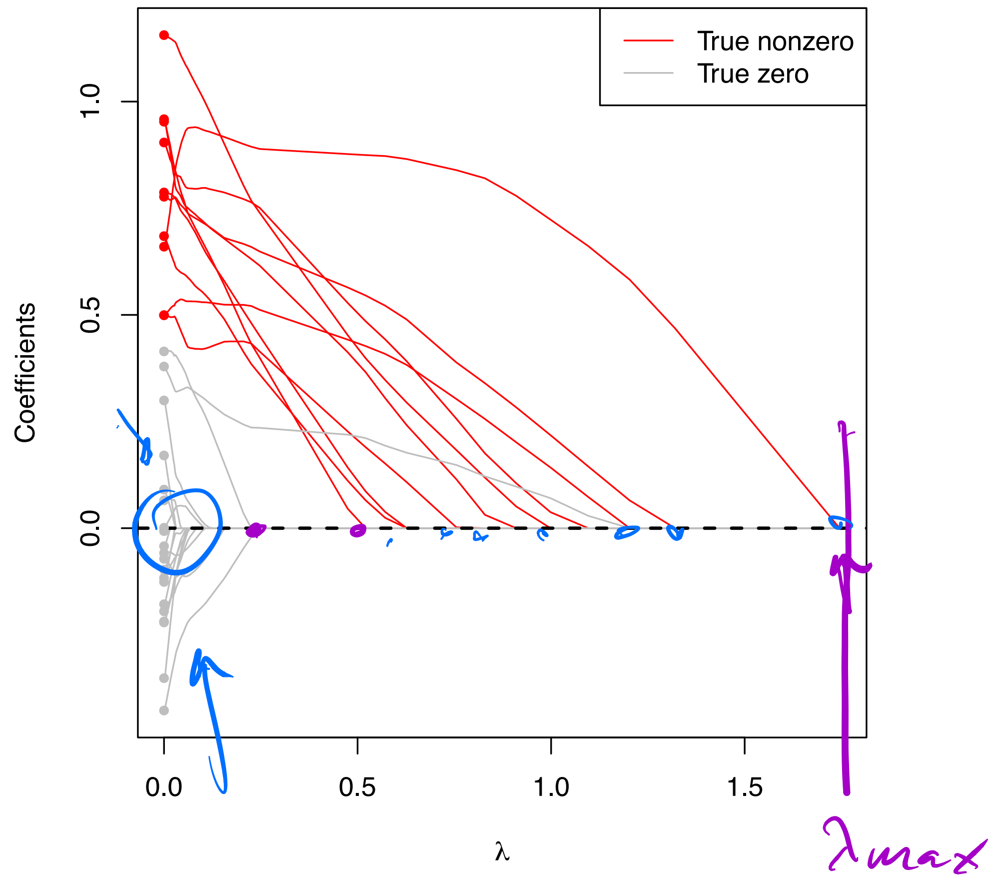
Remember this behavior for Ridge Regression?

Ridge regression coefficients, plotted against λ:



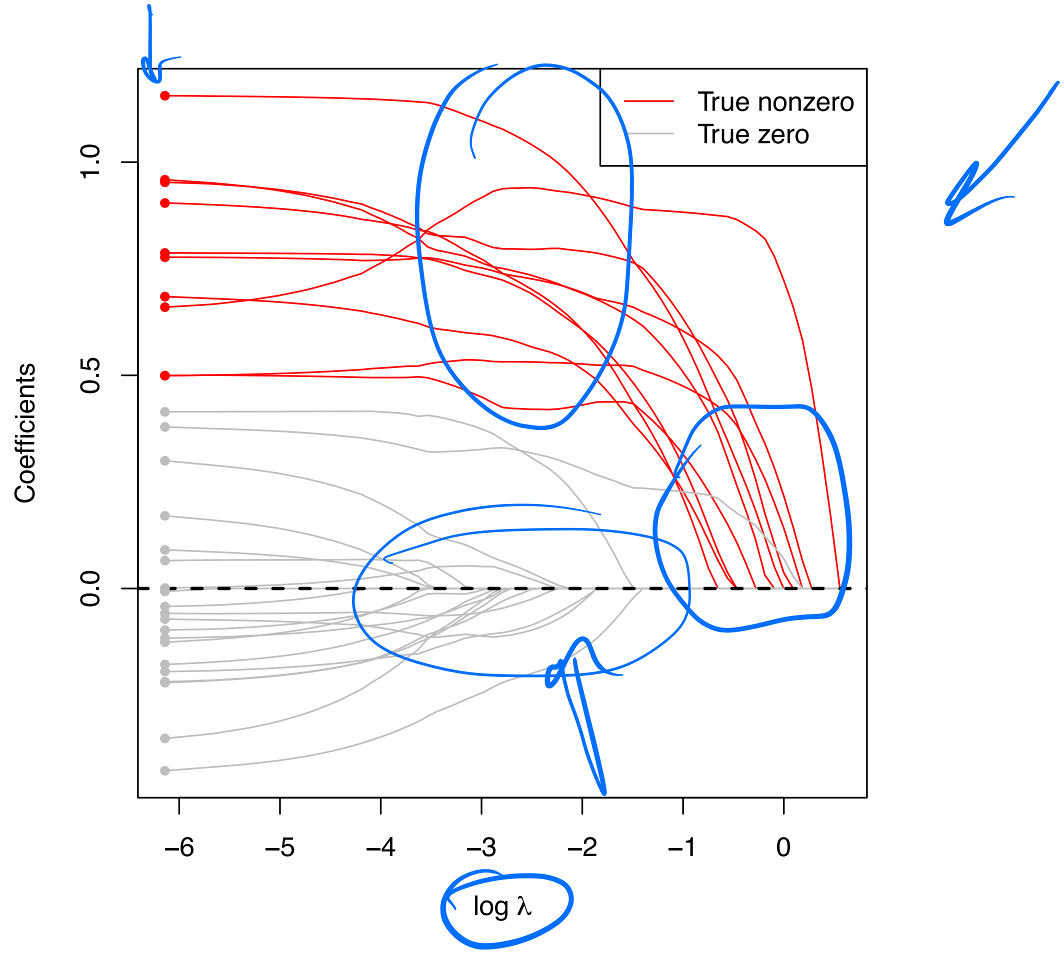
What changes?

Lasso coefficient paths



In the lasso, we get exact zeros! Moreover, the large coefficients are not impacted nearly as much!

Lasso coefficient paths



In the lasso, we get exact zeros! Moreover, the large coefficients are not impacted nearly as much!

It is often easier to view lasso results on the $\log \lambda$ scale.

Advantages of sparsity

- Interpretability: We can understand what the model relies on for prediction (understanding $\hat{f}$)
- We might gain some insight into the underlying data (though not causally) (helping to understand f)
- If we're building a predictive score, we can measure fewer things in the future (simpler $\hat{f}$ to apply later)

Sidenote: Can we do inference (e.g., p-values, confidence intervals, etc)? Not easily!

Important details

Just like in ridge regression:

- We don't penalize our intercept, if one is included. Again, columns are often centered and then intercept omitted.
- The penalty term $\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|$ is not fair if the predictor variables are not on the same scale. Hence, we often scale the columns of X to have variance 1.